

Double Pendulum and The Chaos Theory

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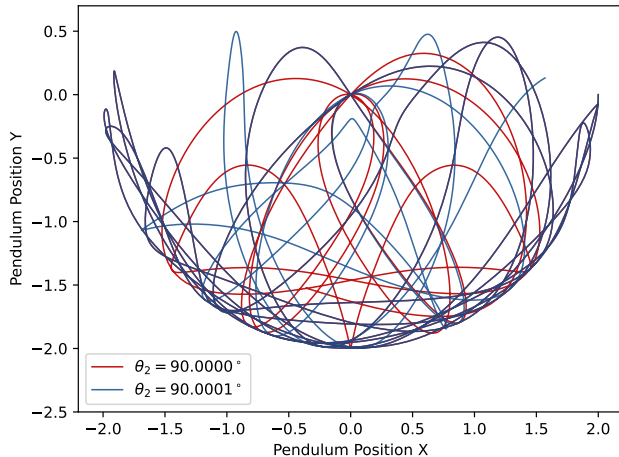
Related CALCAB Knowledge

Differential Equation: It is the core of simulation algorithm we applied in this project, the entire simulation was run based on a differential equation derived by *Lagrangian Mechanics*, so the differential equation we learned in unit 7 is the core of our project.

Limit and Approaching: In our project, the observation of the simulation accuracy as time step changes is also a core part. By speculating the result approaches a value as step decreases (becomes more accurate), we could choose a suitable time step for simulation that is both efficient and accurate.

Introduction To The Chaos Theory

Two Slightly Different Pendula Simulation Outcome



Double Pendulum, Where Newtonian Physics Dies.

Introduction To DE Calculation Approaches

Fourth-Order Runge-Kutta